Answer Key

Grade 6 Mathematics · Unit 5: Area, Volume and Surface Area · Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 5-1

Area of Parallelograms · Area of Rhombuses

Sheet 1 · Area of Parallelograms

1. 11 × 6 = 66 cm² (the 8 cm side is not used)
2. a. 84 ft² b. 81 m² c. 80 in² d. b = 63 ÷ 7 = 9 m
3. One space: 9 × 18 = 162 ft²; 10 spaces: 1,620 ft²
4. h = 108 ÷ 12 = 9 ft (check: 12 × 9 = 108)
5. She multiplied the base by the slanted side (10 × 7). The height must be perpendicular to the base: A = 10 × 5 = 50 cm².

Sheet 2 · Area of Rhombuses

1. (16 × 10) ÷ 2 = 80 cm²
2. a. 70 m² b. 63 ft²
3. 2 × 60 = 120; 120 ÷ 10 = 12 ft
4. A: (12 × 8) ÷ 2 = 48 ft²; B: 8 × h = 48, so h = 6 ft
5. He multiplied the diagonals but did not divide by 2. A = (18 × 8) ÷ 2 = 72 in².

TEST PRACTICE

6. C
7. C – (20 × 9) ÷ 2 = 90 in²

LESSON 5-2

Area of Triangles · Missing Measures in Triangles

Sheet 1 · Area of Triangles

1. ½ × 12 × 7 = 42 cm²
2. ½ × 9 × 8 = 36 in²
3. a. ½ × 10 × 9 = 45 cm² b. ½ × 5 × 11 = 27.5 ft²
4. Area = ½ × 15 × 6 = 45 ft²; 45 ÷ 9 = 5 bags
5. He forgot to take half (40 m² is the parallelogram's area). Correct area: ½ × 8 × 5 = 20 m².

Sheet 2 · Missing Measures in Triangles

1. A: h = 2 × 28 ÷ 8 = 7 in; B: b = 2 × 54 ÷ 9 = 12 ft
2. 2 × 48 = 96; 96 ÷ 16 = 6 m
3. a. 2 × 40 ÷ 10 = 8 m b. 2 × 27 ÷ 6 = 9 in
4. 2 × 35 = 70; 70 ÷ 10 = 7 ft
5. Neither. P = ½ × 12 × 5 = 30 cm² and Q = ½ × 6 × 10 = 30 cm², so the areas are equal.

TEST PRACTICE

6. C
7. B

LESSON 5-3

Trapezoids by Decomposing · The Trapezoid Area Formula

Sheet 1 · Trapezoids by Decomposing

1. a. 6 × 4 = 24 cm² b. triangle base (10 − 6) ÷ 2 = 2 cm; ½ × 2 × 4 = 4 cm² Area: 24 + 4 + 4 = 32 cm²
2. Rectangle 7 × 6 = 42; triangle base 12 − 7 = 5, ½ × 5 × 6 = 15; total 57 in²
3. Rectangle 3 × 4 = 12; triangle base 8 − 3 = 5, ½ × 5 × 4 = 10; total 22 ft²
4. Rectangle 5 × 3 = 15; each triangle base (11 − 5) ÷ 2 = 3, ½ × 3 × 3 = 4.5; total 15 + 4.5 + 4.5 = 24 ft²
5. The two triangle bases together equal 15 − 9 = 6 cm, so each is 3 cm. Rectangle 9 × 4 = 36; triangles 2 × (½ × 3 × 4) = 12; area = 48 cm².

Sheet 2 · The Trapezoid Area Formula

1. ½ × (8 + 14) × 5 = 55 m²
2. a. ½ × 14 × 6 = 42 ft² b. ½ × 14 × 8 = 56 in² c. ½ × 18 × 3 = 27 m² d. ½ × 10 × 9 = 45 cm²
3. ½ × (16 + 24) × 10 = 200 ft²; 200 ÷ 25 = 8 bags
4. 2 × 60 = 120; 120 ÷ 6 = 20 = 8 + b, so b = 12 in

TEST PRACTICE

5. B
6. C

LESSON 5-4

Area of Regular Polygons · Composite Figures

Sheet 1 · Area of Regular Polygons

1. One triangle ½ × 10 × 12.1 = 60.5; 8 × 60.5 ≈ 484 m²
2. Hexagon: P = 24 cm, A ≈ ½ × 24 × 3.5 = 42 cm²; Pentagon: P = 50 ft, A ≈ ½ × 50 × 6.9 = 172.5 ft²
3. a. ≈ ½ × 30 × 4.1 = 61.5 cm² b. ≈ ½ × 60 × 8.7 = 261 ft²
4. Area ≈ 6 × (½ × 6 × 5.2) = 93.6 ft²; cost ≈ 93.6 × $3 = $280.80
5. He found only one of the 5 triangles. Area ≈ 5 × 22 = 110 cm².

Sheet 2 · Composite Figures

1. Bottom 10 × 4 = 40; top 4 × 4 = 16; total 56 m²
2. a. subtract (wall area − window area) b. add (room area + closet area)
3. Top bar 10 × 2 = 20; stem 4 × 4 = 16; total 36 in²
4. Rectangle 12 × 8 = 96; notch base 12 − 3 − 3 = 6, ½ × 6 × 4 = 12; 96 − 12 = 84 cm²
5. 240 − 25 = 215 ft²

TEST PRACTICE

6. A
7. C

LESSON 5-5

Volume of Rectangular Prisms · Missing Fractional Edges

Sheet 1 · Volume of Rectangular Prisms

1. 4 × 3 × 2 = 24 cm³
2. a. 80 in³ b. 400 cm³ c. 216 ft³ d. 315 m³ e. 180 m³ f. 288 in³

Sheet 2 · Missing Fractional Edges

1. 4 × 3/2 × 5/2 = 60/4 = 15 in³
2. a. 24 ft³ b. 28 m³ c. 5/2 × 12 = 30 cm³ d. 9/4 × 4 = 9 in³ e. 4 × ½ × 5 = 10 ft³ f. 6 × 3/2 × 3 = 27 cm³

3. $V = 6 \times 4 \times 1 = 24$ ft³; $24 \div 2 = 12$ bags
4. $B = 20 \times 10 = 200$ in²; $h = 3{,}000 \div 200 = 15$ in
5. No. $A = 6 \times 4 \times 2 = 48$ in³ and $B = 3 \times 4 \times 4 = 48$ in³. They hold the same amount.

3. $3 \times 1\frac{1}{2} = 4\frac{1}{2}$ ft²; $9 \div 4\frac{1}{2} = 2$ ft
4. a. $4 \times 2 \times 3 = 24$ cubes b. $24 \times 1/8 = 3$ in³ (check: $2 \times 1 \times 1\frac{1}{2} = 3$)

TEST PRACTICE

5. D
6. A

LESSON 5-6

Drawing Nets of Solids · Reading Nets and Solids

Sheet 1 · Drawing Nets of Solids

1. Net of 6 rectangles: two 6 in × 2 in, two 6 in × 4 in, two 2 in × 4 in, joined so they fold with no overlaps.
2. a. 6 b. 4 c. 3 d. 5
3. a. $8 \times 8 = 64$ cm² b. $\frac{1}{2} \times 8 \times 6 = 24$ cm² All 4 triangles: 96 cm²
4. two 8 in × 3 in, two 8 in × 12 in, two 3 in × 12 in
5. No. The row wraps around 4 sides, and the last 2 squares overlap the first 2. Nothing covers the top or bottom. A cube net needs squares attached above and below the row.

Sheet 2 · Reading Nets and Solids

1. triangular prism; 5 faces
2. a. rectangular pyramid b. hexagonal prism c. triangular prism d. pentagonal pyramid e. cube f. pentagonal prism
3. a. $6 \times 6 = 36$ in² b. $4 \times (\frac{1}{2} \times 6 \times 5) = 60$ in²
4. Rectangular prism (square prism), 4 cm × 4 cm × 9 cm

TEST PRACTICE

5. C – 3 edges on each triangle + 3 joining edges = 9
6. B – 1 hexagon + 6 triangles

LESSON 5-7

Surface Area of Rectangular Prisms · Surface Area of Triangular Prisms

Sheet 1 · Surface Area of Rectangular Prisms

1. $2(10 \times 4) + 2(10 \times 5) + 2(4 \times 5) = 80 + 100 + 40 = 220$ cm²
2. a. 54 cm² b. 600 ft² c. 28 m² d. 90 in² e. $6 \times 25 = 150$ in² f. $2(18) + 2(12) + 2(6) = 72$ cm²
3. $SA = 2(96) + 2(48) + 2(32) = 352$ in². Yes, 400 > 352, with 48 in² to spare.
4. Top $4 \times 2 = 8$; front and back $2 \times (4 \times 3) = 24$; sides $2 \times (2 \times 3) = 12$; $8 + 24 + 12 = 44$ ft²
5. He found the volume (8 cm³). Surface area adds the 6 faces: $6 \times (2 \times 2) = 24$ cm².

Sheet 2 · Surface Area of Triangular Prisms

1. Triangles $2 \times (\frac{1}{2} \times 6 \times 4) = 24$; rectangles $6 \times 10 + 5 \times 10 + 5 \times 10 = 160$; $SA = 184$ m²
2. a. $2 \times (\frac{1}{2} \times 8 \times 3) = 24$ ft² b. 45 ft²
3. Ends 24; floor $8 \times 9 = 72$; sides $2 \times (5 \times 9) = 90$; total 186 ft²
4. Ends $2 \times (\frac{1}{2} \times 5 \times 12) = 60$; rectangles $5 \times 4 + 12 \times 4 + 13 \times 4 = 120$; $SA = 180$ in²
5. He used the triangle's height (4 m) as a side. The rectangles use the sides of the triangle: 6 × 10, 5 × 10, and 5 × 10.

TEST PRACTICE

6. C
7. B – $6 \times 8 + 8 \times 8 + 10 \times 8 = 192$ cm²

LESSON 5-8

Surface Area of Pyramids · Lateral Area of Pyramids

Sheet 1 · Surface Area of Pyramids

1. $36 + 4 \times (\frac{1}{2} \times 6 \times 9) = 36 + 108 = 144$ m²
2. a. 144 cm² b. $\frac{1}{2} \times 12 \times 10 = 60$ cm² $SA = 144 + 240 = 384$ cm²
3. a. $16 + 4 \times (\frac{1}{2} \times 4 \times 5) = 56$ in² b. $100 + 4 \times (\frac{1}{2} \times 10 \times 12) = 340$ cm²
4. $81 + 4 \times (\frac{1}{2} \times 9 \times 11) = 81 + 198 = 279$ in²
5. He used the pyramid's height (4 ft) instead of the slant height (5 ft). $SA = 36 + 4 \times (\frac{1}{2} \times 6 \times 5) = 36 + 60 = 96$ ft².

Sheet 2 · Lateral Area of Pyramids

1. $4 \times (\frac{1}{2} \times 8 \times 6) = 96$ ft²
2. a. 136 in² b. 72 cm² c. $4 \times (\frac{1}{2} \times 6 \times 10) = 120$ m² d. $4 \times (\frac{1}{2} \times 3 \times 4) = 24$ ft²
3. Lateral area $4 \times (\frac{1}{2} \times 14 \times 10) = 280$ ft²; $280 \div 35 = 8$ bundles
4. One face: $160 \div 4 = 40$ in²; $\frac{1}{2} \times 8 \times$ slant height = 40, so slant height = 10 in
5. No. 85 ft² = 25 + 60 includes the floor. With no floor, only the lateral area is needed: $4 \times (\frac{1}{2} \times 5 \times 6) = 60$ ft².

TEST PRACTICE

6. B
7. C